

Department of Engineering Cybernetics
TTK4215 Adaptive Control
Solution 9

Problem 5 of Chapter 5 in Ioannou & Fidan

a) We write

$$y_p = k_p \frac{Z_p}{R_p} u_p, \quad (1)$$

$$y_m = k_m \frac{Z_m}{R_m} r, \quad (2)$$

where $k_p = b_1$, $Z_p = s + b_0/b_1$, $R_p = s^2 + a_1s + a_0$, $k_m = 4$, $Z_m = 1$, and $R_m = s + 5$. Consider the control law

$$u_p = \frac{\theta_1^*}{\Lambda} u_p + \frac{\theta_2^*}{\Lambda} y_p + \theta_3^* y_p + c_0^* r, \quad (3)$$

where $\Lambda = s + 1$. This control law can be implemented as

$$\dot{\omega}_1 = -\omega_1 + u_p, \quad (4)$$

$$\dot{\omega}_2 = -\omega_2 + y_p, \quad (5)$$

$$u_p = \theta_1^* \omega_1 + \theta_2^* \omega_2 + \theta_3^* y_p + c_0^* r, \quad (6)$$

where $\theta_1^* = (b_1 - b_0)/b_1$, $\theta_2^* = (a_0 - a_1 + 1)/b_1$, $\theta_3^* = (a_1 - 6)/b_1$, and $c_0^* = 4/b_1$ are obtained using the relations (5.81), (5.85) and (5.86) in the book.

b) Let $e = y_p - y_m$, $\theta = [\theta_1, \theta_2, \theta_3, c_0]^T$, and $\omega = [\omega_1, \omega_2, y_p, r]^T$. The control law is then

$$\dot{\omega}_1 = -\omega_1 + u_p, \quad (7)$$

$$\dot{\omega}_2 = -\omega_2 + y_p, \quad (8)$$

$$u_p = \theta^T \omega, \quad (9)$$

where θ is generated by the adaptive law

$$\dot{\theta} = -\Gamma e \omega, \quad (10)$$

where $\Gamma = \Gamma^T > 0$.

c) When it is known that $a_0 = -1$, $a_1 = 0$ and $b_1 = 1$, we see that θ_2^* , θ_3^* , and c_0^* are known. Inserting this information into the control law, we get $u_p = \theta^T \omega = \theta_1 \omega_1 + ((a_0 + a_1 + 1)/b_1) \omega_2 + (a_1 - 6)/b_1 y_p + 4/b_1 r = \theta_1 \omega_1 - 6y_p + 4r$. To implement this, we only need the estimate θ_1 and ω_1 , which are obtained by

$$\dot{\omega}_1 = -\omega_1 + u_p, \quad (11)$$

$$\dot{\theta}_1 = -\gamma e \omega_1, \quad (12)$$

where $\gamma > 0$.

Problem 5 of Chapter 6 in Ioannou & Fidan

a) From the transfer function we have $s^2 y + a s y = s u + b u$. This gives us the parametric model $z = \theta^{*T} \phi$ with

$$z = \frac{s^2}{\Lambda_0(s)} y - \frac{s}{\Lambda_0(s)} u, \quad \theta^* = [a, b]^T, \quad \phi = \left[-\frac{s}{\Lambda_0(s)} y, \frac{1}{\Lambda_0(s)} u \right]^T. \quad (13)$$

$\Lambda_0(s) = (s + \lambda_0)^2$, $\lambda_0 > 0$. Using the gradient algorithm, we have

$$\dot{\theta} = \Gamma \varepsilon \phi, \quad \varepsilon = \frac{z - \theta^T \phi}{m_s^2}, \quad m_s^2 = 1 + \phi^T \phi, \quad \Gamma = \Gamma^T > 0 \quad (14)$$

where $\theta = [\hat{a}, \hat{b}]^T$.

b) Since we are regulating to zero, $Q_m = 1$. Thus, we have $q = 0$, and from $R_p = s^2 + as$, $n = 2$. Thus, L has degree 1, P has degree 1, and A^* has degree 3. So,

$$L = s + l_0, \quad P = p_1s + p_0, \quad (15)$$

and we select

$$A^* = (s + 1)^2. \quad (16)$$

To solve the Diophantine equation, $LQ_mR_p + PZ_p = A^*$, we must require that R_p and Z_p are co-prime. That is $b \neq 0$ and $a \neq b$. Under these conditions, we can solve

$$(s + l_0)(s^2 + as) + (s + b)(p_1s + p_0) = (s + 1)^2$$

to obtain

$$l_0 = \frac{3 - 3b + ab + 1/b}{a - b}, \quad (18)$$

$$p_1 = 3 - a - \frac{3 - 3b + ab + 1/b}{a - b}, \quad (19)$$

$$p_0 = 1/b. \quad (20)$$

Using the estimates from a), we get estimates of the controller parameters as follows

$$\hat{l}_0 = \frac{3 - 3\hat{b} + \hat{a}\hat{b} + 1/\hat{b}}{\hat{a} - \hat{b}}, \quad (21)$$

$$\hat{p}_1 = 3 - \hat{a} - \frac{3 - 3\hat{b} + \hat{a}\hat{b} + 1/\hat{b}}{\hat{a} - \hat{b}}, \quad (22)$$

$$\hat{p}_0 = 1/\hat{b}. \quad (23)$$

The control law becomes

$$u_p = \frac{\lambda - \hat{l}_0}{s + \lambda} u_p - \frac{\hat{p}_1s + \hat{p}_0}{s + \lambda} y_p, \quad (24)$$

where $\lambda > 0$.

c) Loss of stabilizability occurs if $\hat{b} = 0$ or $a - \hat{b} = 0$. Therefore, the stabilizability conditions require that $\hat{b} \neq 0$ and $\hat{a} - \hat{b} \neq 0$ at all times, avoiding zero-pole cancellations in the estimated plant $(s + \hat{b})/(s(s + \hat{a}))$, or equivalently, ensuring that the estimated plant is controllable and observable at all times.

d) The problem can be solved by using projection. This requires that b be bounded away from 0 and that $a - b$ be bounded away from zero. This, in turn, requires some a priori knowledge about the real parameters a and b , so that the initial parameter estimate is on the "right side" of these constraints.